Supplementary Material to "Logic-Based Benders decomposition for additive manufacturing scheduling with sequence-dependent setup times" by Wenjing Ma, Jian Chen, Yaowen Sang, Yingqian Zhang, Yaoxin Wu

S.1 Detailed Computational Results

In the following, we report the detailed results from our computational study.

Table 1 Detailed results of three MIP models

Instances	MIP1			MIP2			MIP3		
	Obj	Gap	Time	Obj	Gap	Time	Obj	Gap	Time
I2_R10_M2-1	35.64	91.58	3600	35.64	0	12.03	35.64	47.59	3600
I2_R10_M2-2	224.98	9.70	3600	224.98	0	0.99	224.98	1.52	3600
I2_R10_M2-3	134.61	0	150.61	134.61	0	1.74	134.61	8.36	3600
I2_R10_M2-4	60.80	91.54	3600	60.80	0	223.82	60.80	53.14	3600
I2_R10_M2-5	72.95	95.06	3600	72.95	0	4.50	72.95	25.97	3600
I2_R10_M3-1	75.55	84.12	3600	75.55	0	2.65	75.55	23.34	3600
I2_R10_M3-2	51.77	68.08	3600	51.77	0	2.79	51.77	17.38	3600
I2_R10_M3-3	69.98	77.48	3600	69.98	0	13.37	69.98	12.00	3600
I2_R10_M3-4	93.49	98.40	3600	80.99	0	243.19	80.99	65.38	3600
I2_R10_M3-5	127.98	78.30	3600	127.31	0	3.65	127.98	14.26	3600
I2_R20_M2-1	233.37	93.49	3600	233.37	0	336.23	233.37	21.71	3600
I2_R20_M2-2	167.39	99.10	3600	167.39	8.29	3600	167.39	41.53	3600

I2_R20_M2-3	97.32	98.97	3600	97.18	9.48	3600	97.15	97.63	3600
I2_R20_M2-4	146.55	92.62	3600	146.55	0	152.07	146.55	30.73	3600
I2_R20_M2-5	121.71	98.77	3600	121.71	7.11	3600	121.68	95.35	3600
I2_R20_M3-1	70.84	98.72	3600	70.84	31.73	3600	70.84	98.75	3600
I2_R20_M3-2	96.06	97.44	3600	96.06	0	1159.02	96.06	34.90	3600
I2_R20_M3-3	87.67	96.19	3600	87.67	0	2411.27	87.67	29.59	3600
I2_R20_M3-4	119.02	98.74	3600	119.02	11.05	3600	119.02	45.83	3600
I2_R20_M3-5	116.02	99.18	3600	116.02	8.90	3600	116.02	63.45	3600
I5_R10_M2-1	21.03	92.87	3600	21.03	0	5.45	21.03	0	745.41
I5_R10_M2-2	46.09	96.87	3600	46.09	0	481.96	46.09	59.45	3600
I5_R10_M2-3	28.31	97.35	3600	28.31	0	2.30	28.31	0	19.42
I5_R10_M2-4	57.51	35.56	3600	57.51	0	2.97	57.51	13.86	3600
I5_R10_M2-5	18.50	59.46	3600	18.50	0	316.16	18.50	8.27	3600
I5_R10_M3-1	23.44	93.60	3600	23.44	0	2.12	23.44	0	25.14
I5_R10_M3-2	28.15	94.67	3600	28.15	0	3.46	28.15	0	151.59
I5_R10_M3-3	61.21	97.55	3600	61.21	0	9.41	61.21	15.61	3600
I5_R10_M3-4	34.91	98.19	3600	34.91	0	136.02	34.91	25.92	3600
I5_R10_M3-5	17.28	96.53	3600	17.28	0	616.27	17.28	59.62	3600
I5_R20_M2-1	31.77	97.55	3600	31.33	21.50	3600	31.33	97.60	3600
I5_R20_M2-2	49.54	100	3600	49.54	2.46	3600	49.54	29.86	3600
I5_R20_M2-3	46.76	98.40	3600	46.76	0	33.85	46.71	98.39	3600
I5_R20_M2-4	40.31	99.07	3600	37.83	19.72	3600	37.83	98.01	3600
I5_R20_M2-5	44.95	96.66	3600	44.95	0	2180.09	44.95	35.36	3600

I5_R20_M3-1	39.81	96.99	3600	39.81	0	2712.67	39.81	64.52	3600
I5_R20_M3-2	70.19	100	3600	70.19	0	2411.91	70.19	46.64	3600
I5_R20_M3-3	45.02	96.67	3600	45.02	0	310.46	45.02	44.69	3600
I5_R20_M3-4	36.03	100	3600	36.03	17.98	3600	35.99	69.80	3600
I5_R20_M3-5	52.33	100	3600	52.13	7.68	3600	52.13	99.04	3600

Table 2 Detailed results of three MIPV models

Instances	MIP1V			MIP2V			MIP3V		
	Obj	Gap	Time	Obj	Gap	Time	Obj	Gap	Time
I2_R10_M2-1	35.64	0	6.16	35.64	0	14.78	35.64	0	16.56
I2_R10_M2-2	224.98	0	1.08	224.98	0	0.46	224.98	0	0.89
I2_R10_M2-3	134.61	0	376.42	134.61	0	3.11	134.61	0	2.71
I2_R10_M2-4	60.80	0	1022.39	60.80	0	21.97	60.80	0	110.32
I2_R10_M2-5	72.95	0	7.78	72.95	0	8.40	72.95	0	2.81
I2_R10_M3-1	75.55	10.25	3600	75.55	0	1.31	75.55	0	1.83
I2_R10_M3-2	51.77	0	3.48	51.77	0	1.43	51.77	0	1.59
I2_R10_M3-3	69.98	0	29.78	69.98	0	7.20	69.98	0	9.08
I2_R10_M3-4	93.49	0	13.15	80.99	0	6.82	80.99	0	5.28
I2_R10_M3-5	127.98	5.56	3600	120.31	0	3.11	120.31	0	6.18
I2_R20_M2-1	233.37	2.82	3600	233.37	0	336.23	233.37	5.78	3600
I2_R20_M2-2	167.39	3.11	3600	167.39	8.29	3600	167.39	23.11	3600
I2_R20_M2-3	97.17	2.28	3600	97.18	8.87	3600	97.15	35.28	3600
I2_R20_M2-4	146.55	3.65	3600	146.55	0	152.07	146.55	9.33	3600

I2_R20_M2-5	121.71	0	15.99	121.71	6.74	3600	121.68	36.72	3600
I2_R20_M3-1	70.84	1.29	3600	70.84	23.98	3600	70.84	46.62	3600
I2_R20_M3-2	96.06	5.09	3600	96.06	0	1826.72	96.06	15.84	3600
I2_R20_M3-3	87.67	9.97	3600	87.67	0	1713.63	87.67	21.35	3600
I2_R20_M3-4	119.02	7.80	3600	119.02	10.86	3600	119.02	21.75	3600
I2_R20_M3-5	116.02	2.98	3600	116.02	7.93	3600	116.02	13.62	3600
I5_R10_M2-1	21.03	0	2.92	21.03	0	5.45	21.03	0	1.78
I5_R10_M2-2	46.09	0	3.10	46.09	0	481.96	46.09	0	6.71
I5_R10_M2-3	28.31	0	2.03	28.31	0	2.30	28.31	0	1.87
I5_R10_M2-4	57.51	0	3.12	57.51	0	2.97	57.51	0	0.94
I5_R10_M2-5	18.50	0	4.10	18.50	0	316.16	18.50	0	29.27
I5_R10_M3-1	23.44	0	2.88	23.44	0	2.12	23.44	0	1.47
I5_R10_M3-2	28.15	0	1.99	28.15	0	3.46	28.15	0	2.85
I5_R10_M3-3	61.21	0	18.32	61.21	0	9.41	61.21	0	2.24
I5_R10_M3-4	34.91	0	11.99	34.91	0	136.02	34.91	0	6.65
I5_R10_M3-5	17.28	0	4.61	17.28	0	616.27	17.28	0	7.30
I5_R20_M2-1	31.77	21.50	3600	31.33	0	812.78	31.33	57.17	3600
I5_R20_M2-2	49.54	2.46	3600	49.54	0	212.87	49.54	18.82	3600
I5_R20_M2-3	46.76	0	33.85	46.76	0	44.08	46.71	0	1081.24
I5_R20_M2-4	37.83	19.72	3600	37.83	0	48.40	37.83	27.21	3600
I5_R20_M2-5	44.95	0	62.51	44.95	0	68.05	44.95	0	766.67
I5_R20_M3-1	39.81	0	2712.67	39.81	0	140.13	39.81	9.20	3600
I5_R20_M3-2	70.19	0	2411.91	70.19	0	266.58	70.19	16.66	3600

I5_R20_M3-3	45.02	0	310.46	45.02	1.47	3600	45.02	24.70	3600
I5_R20_M3-4	36.03	17.98	3600	36.03	0	998.72	35.99	35.14	3600
I5_R20_M3-5	52.13	7.68	3600	52.13	0	630.86	52.13	18.70	3600

Table 3 Detailed results of MIP2V and LBBD of small-sized instances

Instances	MIP2V			LBBD		
	Obj	Gap	Time	Obj	Gap	Time
I2_R10_M2-1	35.64	0	14.78	35.64	0	0.24
I2_R10_M2-2	224.98	0	0.46	224.98	0	0.07
I2_R10_M2-3	134.61	0	3.11	134.61	0	0.27
I2_R10_M2-4	60.80	0	21.97	60.80	0	1.77
I2_R10_M2-5	72.95	0	8.40	72.95	0	0.31
I2_R10_M3-1	75.55	0	1.31	75.55	0	0.29
I2_R10_M3-2	51.77	0	1.43	51.77	0	0.11
I2_R10_M3-3	69.98	0	7.20	69.98	0	0.15
I2_R10_M3-4	80.99	0	6.82	91.63	0	0.60
I2_R10_M3-5	120.31	0	3.11	127.98	0	1.03
I2_R20_M2-1	233.37	0	336.23	233.37	0	4.63
I2_R20_M2-2	167.39	8.29	3600	167.22	0.67	3600
I2_R20_M2-3	97.18	8.87	3600	97.18	2.28	3600
I2_R20_M2-4	146.55	0	152.07	146.55	0	11.62
I2_R20_M2-5	121.71	6.74	3600	121.71	0	3093.30
I2_R20_M3-1	70.84	23.98	3600	70.84	0	99.78

I2_R20_M3-2	96.06	0	1826.72	96.06	0	713.42
I2_R20_M3-3	87.67	0	1713.63	87.67	0	43.07
I2_R20_M3-4	119.02	10.86	3600	119.02	0	121.32
I2_R20_M3-5	116.02	7.93	3600	116.02	2.74	3600
I3_R10_M2-1	20.14	0	2.89	20.14	0	0.53
I3_R10_M2-2	50.61	0	2.77	50.61	0	0.28
I3_R10_M2-3	35.84	0	35.50	35.84	0	2.11
I3_R10_M2-4	31.20	0	10.06	31.21	0	3.76
I3_R10_M2-5	39.38	0	0.06	39.38	0	0.04
I3_R10_M3-1	32.62	0	2.54	32.62	0	0.31
I3_R10_M3-2	40.29	0	0.34	40.29	0	0.02
I3_R10_M3-3	27.71	0	46.47	27.71	0	6.71
I3_R10_M3-4	47.91	0	2.71	47.91	0	0.43
I3_R10_M3-5	17.08	0	6.69	17.08	0	0.31
I3_R20_M2-1	43.56	12.85	3600	43.56	0	37.80
I3_R20_M2-2	73.40	16.24	3600	73.40	0	961.09
I3_R20_M2-3	56.76	20.74	3600	56.76	2.09	3600
I3_R20_M2-4	54.95	23.97	3600	54.95	0	43.02
I3_R20_M2-5	70.71	0	9.05	70.71	0	3.01
I3_R20_M3-1	55.09	29.78	3600	55.09	0	30.08
I3_R20_M3-2	76.97	40.30	3600	76.97	0	23.97
I3_R20_M3-3	45.45	0	1014.74	45.45	0	8.04
I3_R20_M3-4	68.81	40.78	3600	68.81	7.54	3600

I3_R20_M3-5	66.76	25.01	3600	66.76	0	76.89
I5_R10_M2-1	21.03	0	5.45	21.03	0	0.21
I5_R10_M2-2	46.09	0	481.96	46.09	0	1.45
I5_R10_M2-3	28.31	0	2.30	28.31	0	0.19
I5_R10_M2-4	57.51	0	2.97	57.51	0	0.12
I5_R10_M2-5	18.50	0	316.16	18.50	0	4.55
I5_R10_M3-1	23.44	0	2.12	23.44	0	0.07
I5_R10_M3-2	28.15	0	3.46	28.15	0	0.16
I5_R10_M3-3	61.21	0	9.41	61.21	0	0.29
I5_R10_M3-4	34.91	0	136.02	34.91	0	0.37
I5_R10_M3-5	17.28	0	616.27	17.28	0	0.69
I5_R20_M2-1	31.33	21.50	3600	31.33	0	32.70
I5_R20_M2-2	49.54	2.46	3600	49.54	0	22.02
I5_R20_M2-3	46.76	0	33.85	46.76	0	2.07
I5_R20_M2-4	37.83	19.72	3600	37.83	0	13.06
I5_R20_M2-5	44.95	0	2180.09	44.95	0	1.19
I5_R20_M3-1	39.81	0	2712.67	39.81	0	1.98
I5_R20_M3-2	70.19	0	2411.91	70.19	0	1.32
I5_R20_M3-3	45.02	0	310.46	45.02	0	1.11
I5_R20_M3-4	36.03	17.98	3600	36.03	0	111.05
I5_R20_M3-5	52.13	7.68	3600	52.13	0	9.83

Table 4 Detailed results of MIP2V and LBBD of medium-sized instances

	MIP2V			LBBD		
	Obj	Gap	Time	Obj	Gap	Time
I2_R30_M2-1	105.36	14.48	3600	104.32	15.33	3600
I2_R30_M2-2	193.78	25.82	3600	193.28	18.37	3600
I2_R30_M2-3	83.08	22.22	3600	82.13	15.04	3600
I2_R30_M2-4	92.02	10.75	3600	89.36	13.33	3600
I2_R30_M2-5	108.84	13.72	3600	108.60	18.29	3600
I2_R30_M3-1	98.15	27.54	3600	97.47	27.04	3600
I2_R30_M3-2	198.24	30.06	3600	198.24	0	3600
I2_R30_M3-3	92.02	15.00	3600	91.51	5.27	3600
I2_R30_M3-4	133.47	33.86	3600	133.03	16.24	3600
I2_R30_M3-5	328.80	12.85	3600	328.80	11.52	3600
I2_R40_M2-1	137.79	17.62	3600	137.19	17.25	3600
I2_R40_M2-2	261.68	33.09	3600	260.75	32.85	3600
I2_R40_M2-3	68.56	3.24	3600	67.86	2.24	3600
I2_R40_M2-4	112.09	12.73	3600	110.80	11.71	3600
I2_R40_M2-5	394.36	1.47	3600	392.92	1.11	3600
I2_R40_M3-1	169.46	27.56	3600	165.45	25.80	3600
I2_R40_M3-2	160.94	36.64	3600	160.83	36.60	3600
I2_R40_M3-3	116.61	25.47	3600	113.22	23.64	3600
I2_R40_M3-4	243.36	5.33	3600	239.58	3.84	3600
I2_R40_M3-5	171.59	35.11	3600	171.30	21.08	3600

I3_R30_M2-1	69.98	22.33	3600	69.98	4.68	3600
I3_R30_M2-2	119.08	14.16	3600	118.83	10.77	3600
I3_R30_M2-3	79.70	19.13	3600	78.45	13.54	3600
I3_R30_M2-4	61.84	16.55	3600	61.50	16.10	3600
I3_R30_M2-5	144.07	15.07	3600	142.75	5.10	3600
I3_R30_M3-1	67.30	33.05	3600	67.07	32.83	3600
I3_R30_M3-2	90.54	18.72	3600	89.61	17.88	3600
I3_R30_M3-3	88.17	19.42	3600	88.05	13.37	3600
I3_R30_M3-4	126.77	15.59	3600	126.77	3.97	3600
I3_R30_M3-5	289.49	7.24	3600	289.49	0	62.24
I3_R40_M2-1	112.59	30.00	3600	110.80	28.87	3600
I3_R40_M2-2	139.59	23.49	3600	135.83	21.37	3600
I3_R40_M2-3	48.31	12.13	3600	42.45	0	51.16
I3_R40_M2-4	79.39	11.88	3600	77.53	9.77	3600
I3_R40_M2-5	371.08	7.24	3600	371.08	0.06	3600
I3_R40_M3-1	335.23	16.15	3600	335.23	0	46.46
I3_R40_M3-2	75.03	26.04	3600	73.93	24.95	3600
I3_R40_M3-3	100.95	14.86	3600	100.95	0.02	3600
I3_R40_M3-4	98.54	33.76	3600	98.53	31.17	3600
I3_R40_M3-5	120.19	33.36	3600	120.07	33.29	3600
I5_R30_M2-1	47.41	14.40	3600	47.41	5.44	3600
I5_R30_M2-2	81.98	9.42	3600	81.52	7.06	3600
I5_R30_M2-3	57.12	0	1461.39	57.12	0.00	8.38

I5_R30_M2-4	49.81	21.38	3600	49.45	12.30	3600
I5_R30_M2-5	72.63	40.57	3600	72.42	14.61	3600
I5_R30_M3-1	52.91	36.67	3600	52.47	24.13	3600
I5_R30_M3-2	78.93	42.85	3600	77.85	28.71	3600
I5_R30_M3-3	53.02	16.13	3600	52.92	14.87	3600
I5_R30_M3-4	50.65	42.82	3600	48.16	26.40	3600
I5_R30_M3-5	176.78	17.89	3600	176.78	0	21.58
I5_R40_M2-1	81.90	41.47	3600	75.99	27.99	3600
I5_R40_M2-2	74.62	19.79	3600	73.20	18.24	3600
I5_R40_M2-3	128.16	10.46	3600	128.16	0	8.28
I5_R40_M2-4	72.70	32.49	3600	70.75	20.76	3600
I5_R40_M2-5	277.99	7.41	3600	277.99	0	31.05
I5_R40_M3-1	129.60	43.53	3600	128.93	3.48	3600
I5_R40_M3-2	63.37	32.56	3600	60.92	29.85	3600
I5_R40_M3-3	104.82	14.31	3600	104.82	0	13.79
I5_R40_M3-4	63.46	36.13	3600	61.58	32.84	3600
I5_R40_M3-5	86.11	23.86	3600	84.77	17.35	3600

Table 5 Detailed results of MIP2V and LBBD of large-sized instances

	MIP2V			**LBBD**		
	Obj	Gap	Time	Obj	Gap	Time
I5_R50_M3-1	111.78	21.10	3600	106.69	17.33	3600
I5_R50_M5-2	89.89	33.76	3600	88.52	32.73	3600

I5_R50_M3-3	47.59	21.96	3600	44.41	16.39	3600
I5_R50_M3-4	124.47	27.41	3600	123.66	25.12	3600
I5_R50_M3-5	83.21	39.22	3600	80.41	37.10	3600
I5_R50_M5-1	93.11	46.54	3600	89.66	39.44	3600
I5_R50_M5-2	96.36	37.49	3600	96.36	0.84	3600
I5_R50_M5-3	74.55	39.51	3600	73.31	38.48	3600
I5_R50_M5-4	123.36	51.52	3600	119.83	41.56	3600
I5_R50_M5-5	110.35	43.78	3600	106.57	41.79	3600
I5_R80_M3-1	183.55	31.17	3600	180.39	29.97	3600
I5_R80_M3-2	300.30	49.54	3600	291.29	32.80	3600
I5_R80_M3-3	134.13	41.12	3600	128.17	21.27	3600
I5_R80_M3-4	152.97	40.27	3600	150.68	39.36	3600
I5_R80_M3-5	120.74	28.40	3600	119.28	27.52	3600
I5_R80_M5-1	371.48	59.76	3600	362.92	37.76	3600
I5_R80_M5-2	133.93	28.71	3600	128.87	25.91	3600
I5_R80_M5-3	131.94	31.88	3600	131.87	31.85	3600
I5_R80_M5-4	161.77	28.63	3600	151.29	23.69	3600
I5_R80_M5-5	169.15	39.44	3600	166.40	38.44	3600
I5_R100_M3-1	224.80	14.01	3600	222.55	7.40	3600
I5_R100_M3-2	637.40	68.20	3600	634.11	39.61	3600
I5_R100_M3-3	192.94	51.43	3600	187.69	9.59	3600
I5_R100_M3-4	175.82	31.28	3600	163.57	26.14	3600
I5_R100_M3-5	207.23	35.69	3600	203.92	34.65	3600

I5_R100_M5-1	208.43	44.01	3600	199.23	41.43	3600
I5_R100_M5-2	174.66	37.88	3600	165.06	26.22	3600
I5_R100_M5-3	184.81	42.32	3600	181.50	41.27	3600
I5_R100_M5-4	266.56	50.94	3600	252.25	48.16	3600
I5_R100_M5-5	175.94	30.02	3600	175.53	29.85	3600
I10_R50_M3-1	44.87	47.48	3600	44.25	20.70	3600
I10_R50_M3-2	47.44	30.85	3600	46.97	30.15	3600
I10_R50_M3-3	25.47	29.92	3600	25.41	29.76	3600
I10_R50_M3-4	44.38	16.20	3600	44.38	0	30.72
I10_R50_M3-5	37.52	31.38	3600	37.35	31.08	3600
I10_R50_M5-1	57.31	49.79	3600	53.83	31.28	3600
I10_R50_M5-2	51.39	38.36	3600	51.10	30.51	3600
I10_R50_M5-3	55.23	49.41	3600	55.03	31.89	3600
I10_R50_M5-4	55.15	42.18	3600	56.47	28.59	3600
I10_R50_M5-5	51.10	46.61	3600	50.22	34.52	3600
I10_R80_M3-1	81.63	42.29	3600	74.71	24.06	3600
I10_R80_M3-2	82.63	47.54	3600	75.66	29.87	3600
I10_R80_M3-3	70.25	49.17	3600	67.02	41.65	3600
I10_R80_M3-4	75.03	44.13	3600	68.03	33.30	3600
I10_R80_M3-5	67.56	39.79.21	3600	62.16	35.21	3600
I10_R80_M5-1	105.30	25.49	3600	96.04	20.03	3600
I10_R80_M5-2	65.55	33.27	3600	60.48	21.22	3600
I10_R80_M5-3	85.61	49.79	3600	76.94	36.19	3600

I10_R80_M5-4	92.34	46.16	3600	89.85	12.27	3600
I10_R80_M5-5	63.83	46.55	3600	59.94	37.00	3600
I10_R100_M3-1	112.55	53.78	3600	111.05	9.14	3600
I10_R100_M3-2	100.14	46.97	3600	96.55	45.00	3600
I10_R100_M3-3	76.31	38.48	3600	69.48	32.43	3600
I10_R100_M3-4	105.63	44.90	3600	93.30	37.62	3600
I10_R100_M3-5	80.38	35.83	3600	75.44	31.63	3600
I10_R100_M5-1	91.09	38.33	3600	85.41	34.23	3600
I10_R100_M5-2	85.51	38.71	3600	76.90	31.85	3600
I10_R100_M5-3	107.07	45.04	3600	103.81	43.31	3600
I10_R100_M5-4	156.52	58.87	3600	155.80	32.64	3600
I10_R100_M5-5	93.09	47.21	3600	84.20	41.64	3600